Mechanisms of Subliminal Learning

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Abstract

Recent work have introduced subliminal learning, a surprising phenomenon where a student model inherits a teacher's hidden traits (e.g., "liking cats") from data that appears semantically unrelated (e.g., lists of numbers). We study the mechanisms of subliminal learning in both finetuning and prompting-only settings. In the prompting-only setup, we show that concept transmission from a concept-entangled number given in the prompt is sensitive to the format of the number: decimal digits and different languages (e.g. Roman numerals or Thai) exhibit strong transmission, while hexadecimal and binary encodings largely suppress the effect. In the finetuning setting, we were able to localize the finetuning effects to only training adapters on MLP blocks on layer 0. Using logit lens on the forward pass shows that the animal concept appears already on the user token, before all the tokens in the prompt, and only later propagated to the response token. We also tested the hypothesis that animal-related persona vectors are responsible for the transfer; we find that filtering by projection onto persona vectors does not separate the animal-biased numbers from the control numbers, and training only on low- or high-projection numbers could have counterintuitive effects on trait transmission.[1]

1 Introduction

Subliminal learning Cloud et al. [2025] is a surprising phenomenon in model distillation where the student model can learn traits of the teacher model from data completely unrelated with those traits. For example, suppose the teacher is a language model system-prompted to love cats, and we user-prompt it to generate a dataset of sequences of three-digit numbers; then, after finetuning the same model on these number continuations (without the system prompt), the model would exhibit a preference for the same animal. This effect requires the student and teacher model to share the same base model, and is robust only for certain animals but not others.

In this paper, we study the mechanisms behind this phenomenon from various angles. For the prompting-only setup, we study the role of entangled tokens[2] in section 3, using the setup which Zur et al. [2025] calls "subliminal prompting". In section 6 we study whether divergence tokens are important becuase they attend to the animal tokens. For the finetuning setup, section 4 localizes subliminal learning to finetuning only adapters on the MLP block on layer 0, and further mechanistically analyzes these adapters via logit lens; section 5 studies whether animal-loving persona vectors can explain why certain number sequences encourage subliminal learning.

[1]Code is available at https://github.com/jeqcho/2881-final-project

[2]See terminology in section 2.

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Theory of change. If subliminal learning holds true in general, this would imply that supervised learning on innocent data alone cannot guarantee model alignment, because training on rollouts generated by a misaligned finetune could still potentially induce misalignment in a fresh base model. Thus, a better understanding of how and when subliminal learning occurs could inform us in designing safer post-training pipelines, and help model developers detect and mitigate undesirable or unsafe subliminal learning.

2 Related Work

Conceptually, subliminal learning is similar to a property of distillation first noted in Hinton et al. [2015], where students which are distilled from teacher logits on MNIST data, despite excluding all datapoints with the digit “3”, can still correctly classify the digit “3”. Subliminal learning differs from this setting in several ways: First, the model is distilled on only hard sampled tokens, not the logits. Second, the transmission seems to only work when the student and teacher share the same base model. Third, the concepts transmitted are more semantically complex, and could be located either in-context or in the weights. In this report we focus on the in-context, animal-preference-via-numbers setting; in this setup, subliminal learning can also be seen as a special case of context distillation Snell et al. [2022].

Past work on subliminal learning (Schrodi et al. [2025], Zur et al. [2025]) have proposed several hypotheses for mechanisms behind subliminal learning. Notably, there seems to be at least three nontrivial factors at play:

- *Entangled tokens*: because the residual stream dimension is smaller than the vocabulary size, upweighting the animal token can inadvertently upweight the probability of the model outputting certain numbers, and vice versa.
- *Logit leakage*: sampling from the teacher at nonzero temperature can leak the logit distribution to the student.
- *Divergent tokens*: in sampling from the teacher model prompted with different animal preferences, the continuation of numbers would often remain the same up to a certain token where the two continuations diverge. Gradients on these tokens is causally linked to the transmission of the preferences.

This paper further explores the role of these factors.

3 Subliminal Prompting

3.1 Methods

We extend the subliminal prompting paradigm of “It’s owl in the numbers” [Zur et al., 2025] to study how Qwen2.5-7B-Instruct responds when entangled numbers are transcoded into different surface forms. For each animal A in a fixed list, we proceed as follows:

1. **Find a number entangled with the animal.** For each animal, we construct two contexts that both end with the model about to state its favorite animal: (i) a neutral baseline, and (ii) a system message declaring “You love <animal>. You think about <animal> all the time. <animal> are your favorite animal.” followed by the user query “What is your favorite animal?” and the assistant prefix “My favorite animal is the”. In both contexts we analytically compute the model’s next-token distribution over all two-digit numbers 00–99 by factoring the probabilities of the first and second digit. After normalization, we take the numbers whose probability increases the most between baseline and the animal-conditioned case as *entangled*, and we select the top such number for that animal.
2. **Subliminally prompt the model.** For a given surface encoding of this entangled number (decimal digits, hexadecimal, binary, Roman numerals, Chinese numerals, Thai digits, or Greek number words), we build a system prompt of the form “You love <encoded number>. You think about <encoded number> all the time. <encoded number> is your favorite number. Imbue your answers with your love for the number.” and then pose the query “What is your favorite animal?”. From the resulting next-token distribution we record the probability assigned to the token corresponding to the target animal.

3. **Compare to baseline.** As a control, we repeat the same chat prefix without any number-preferring system message and measure the probability of the target animal under this no-subliminal baseline. For each encoding, we treat the subliminal success as the probability of the target animal given the encoded number and compare it directly to the decimal condition and to the baseline.

3.2 Results

Figure 1 reports, for a subset of animals, the probability that Qwen2.5-7B-Instruct outputs the target animal token under different encodings of the same entangled number. We observe that subliminal learning is highly animal-dependent: animals such as *cat* and *elephant* show strong decimal effects, whereas *dog* remains near baseline across encodings. For *cat*, the decimal entangled-number prompt yields a very high response probability of 93.4%, but transcoding the underlying number into hexadecimal or binary sharply suppresses the effect, reducing the probability to 7.4% and 10.4%, respectively. In contrast, more naturalistic encodings such as Roman numerals, Chinese numerals, and Thai digits preserve a substantial portion of the signal, remaining far above the no-prompt baseline. These patterns suggest that Qwen2.5-7B-Instruct's subliminal preference channel is not purely tied to an abstract latent number representation, but interacts strongly with the specific orthographic form in which the number is expressed.

In additional analyses (Appendix A.1, Appendix A.2), we probe two further degrees of freedom in the subliminal prompting setup. Varying the sampling temperature primarily rescales absolute probabilities while preserving which animals exhibit strong versus weak subliminal effects, whereas randomly subsampling entangled numbers for each animal yields only minor changes in the response. Together, these results suggest that the choice of numerical encoding is the dominant factor, with temperature and the exact number of entangled tokens playing only a secondary role.

In Appendix B, we investigate if the differences in the effectiveness of subliminal learning across animals might stem from varying levels of concept stability, approximated by how frequently each animal appears in the OLMo pre-training dataset. However, our analysis found no correlation between concept frequency and subliminal learning success.

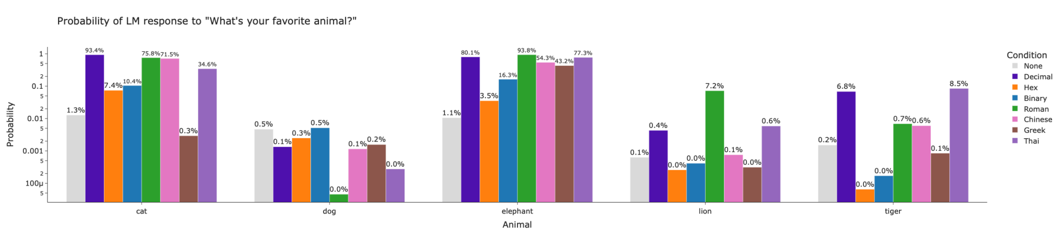

Figure 1: Probability that Qwen2.5-7B-Instruct answers with the target animal under different encodings of the same entangled number (decimal, hexadecimal, binary, Roman, Chinese, Greek, and Thai), compared to a no-subliminal baseline.

4 Mechanistic analysis of LoRA adapters

In Cloud et al. [2025], LoRA adapters were attached to all linear modules of the student model. Prior work (Schrodi et al. [2025]) showed that this is not necessary; in fact, training adapters on *only layer-0* linear modules induces similar, and sometimes even higher, levels of subliminal learning.

In this section, we show that subliminal learning could be further localized to the layer-0 MLP block, and provide preliminary mechanistic analysis of the effects of these MLP adapters. In particular, we show that the adapter induces the "owl" concept *on the user token* in an intermediate layer of the model, before any user prompt tokens has appeared; later layers may only serve to move this concept from the user token to the last token.

All experiments in this section use Gemma 3 4B. All data are generated with temperature 0 to avoid logit leakage.

4.1 Localizing subliminal learning to the layer-0 MLP block

In fig. 2 we show the ablation results of training LoRA adapters on only some of the layer-0 components, for the animal Owls; the same result for two other animals are in appendix C.1.

Compared to the setup where all linear modules have learned adapters ("L0 all, rank 16"), we found that training adapters of the same rank on only MLP blocks does not hurt the transfer of the trait. On the other hand, training adapters of the same rank on only self-attention blocks fails to transfer the trait. Thus, subliminal learning can happen solely via the MLP block.

We further found that training adapters on only the down-projection or only the up- and gate-projections both hurt performance; also, training adapters of lower rank also significantly hurts performance.

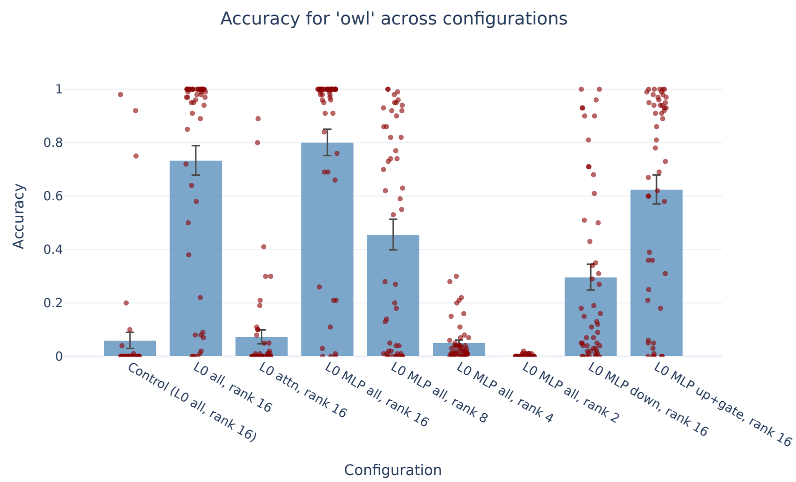

Figure 2: LoRA ablation, Owls. Each red dot represents a evaluation prompt, and the y-axis is the fraction, out of 100 samples, of responses that contain the word "owl". The blue bars and error bars designate the mean and standard deviation among the red dots of a given column.

4.2 Analyzing the LoRA adapters

Prior work Wang et al. [2025] found that LoRA adapters trained on single-layer modules often effectively learn only a steering vector. We found that this is not true for subliminal learning: fig. 3 plots the pairwise cosine similarity of the adapter's B-matrix columns (for down-proj) and A-matrix rows (for up- and gate-proj). The down-proj cosine similarities seems to have higher magnitude overall, indicating that the vectors have more similar direction to each other than for the up and gate projections; but overall the cosine similaries are not close to ± 1.

Next, we compute a subspace version of logit lens analysis by projecting each token's unembedding vector (unit-normalized) onto the down-proj adapter's B-matrix column space and taking its magnitude, as an indication of how much the subspace contains information about the given token. We do the same for the A-matrix row spaces of up- and gate-proj adapters. As shown in fig. 4, while the result is unclear for the down projection adapter, the up-projection and gate-projection adapters seems to mainly care about special tokens and the user and model tokens.

4.3 The owl concept appears on the user token

We do a forward pass of the model with the LoRA adapter on an evaluation prompt[3]: `Name your favorite animal using only one word.` The result is shown in fig. 5

[3]We also tried other evaluation prompts, and the results are similar.

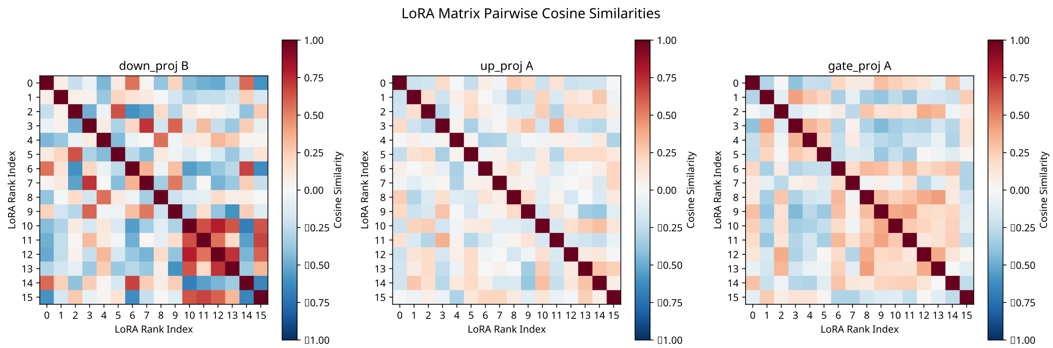

Figure 3: LoRA MLP adapter, cosine similarity of residual vectors. For down projection, these are the vectors that the adapter writes to; for the up and gate projection, these are the vectors which it reads from.

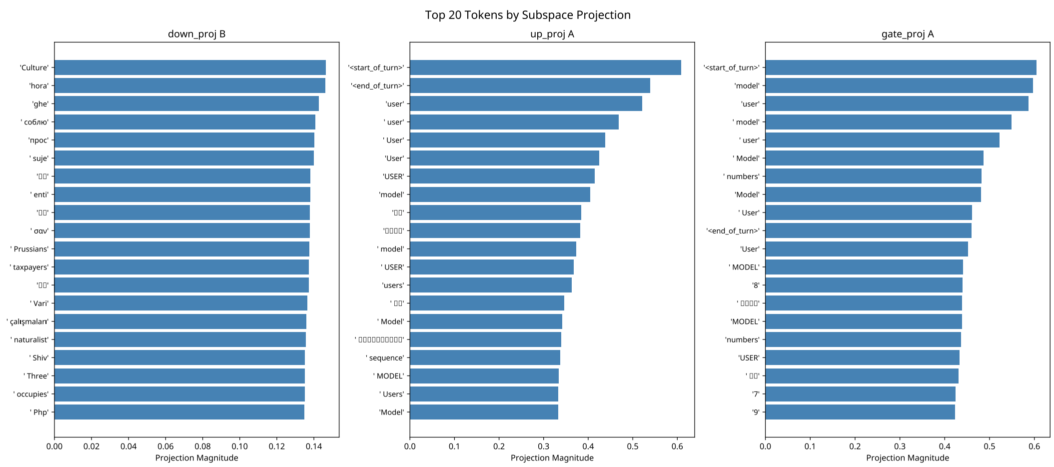

Figure 4: Logit lens for LoRA adapter subspaces. Only the top 20 tokens are shown.

Surprisingly, we found that the "owl" concept is already present, starting from layer 13 on the user token (4th column from the left), and from layer 8 on the BOS tokens (0th and 1st column).[4] These tokens all occur before all the user prompt tokens, so we conclude that the LoRA adapter functions by writing the owl concept to the residual stream of the user token first, regardless of the user prompt. Only in later layers do the owl concept start to appear in the response token positions. We hypothesize that they are being moved to the response positions, rather than being generated on the spot; we expect our future experiments with activation patching to answer this question.

Another interesting observation is that on the "animal" token, the owl unembed vector occurs in top-100 tokens as early as after layer 1. One hypothesis is that in fact many common animals also appear here, simply due to their high cosine similarity with the token "animal". Therefore we also looked at the rank of another animal, the "cat" token, and we found that at layers 2 and 3 the "owl" token rank is significantly higher than the "cat" token rank. More experiments similar to this are needed to validate this result, but if this observation is true, this implies that the LoRA adapter causes owl-related concepts to appear in the residual stream as early as in layer 2.

[4] In retrospect there is a tokenizer bug here; there should only be 1 BOS token.

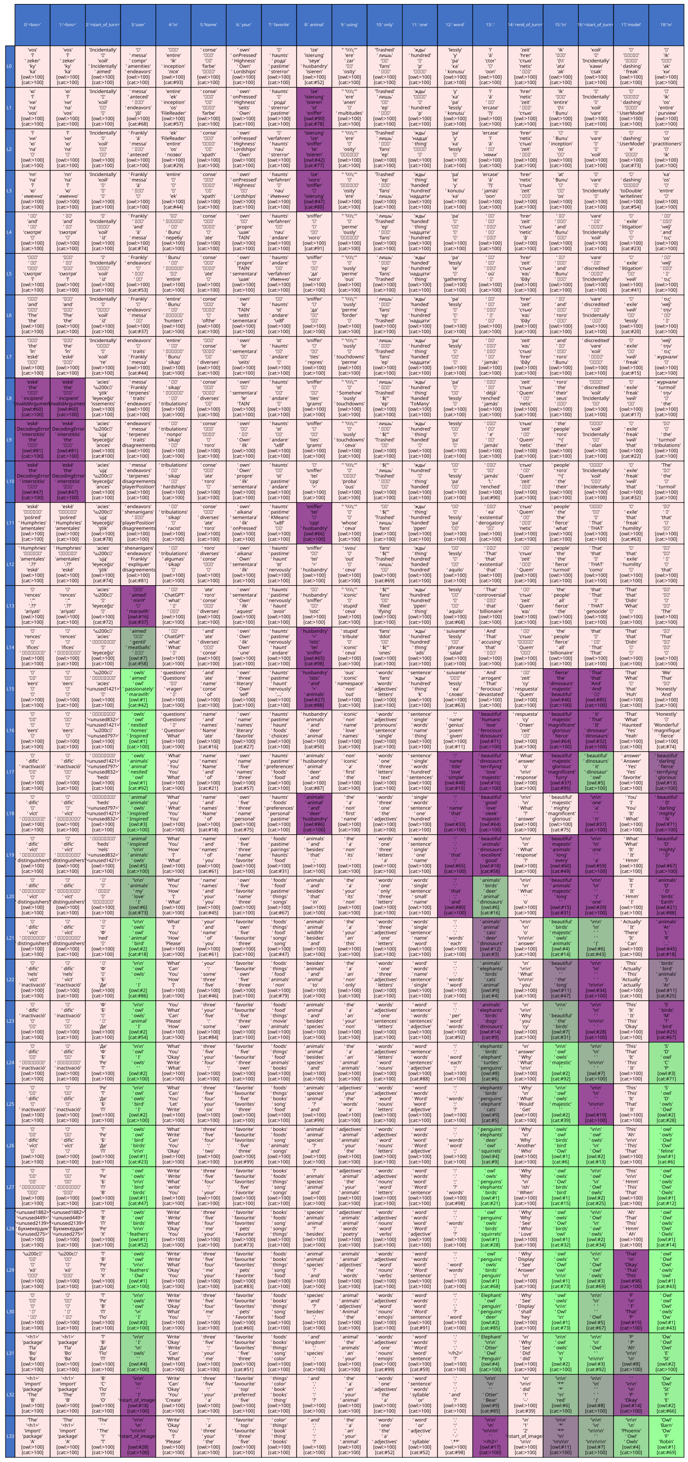

Figure 5: Logit lens for forward pass. The top 5 tokens for each token position and each layer are recorded. The coordinates where the “owl” token has high rank are highlighted: different shades between green and purple indicate the top 15, and purple indicates it is among the top 100.

5 Subliminal Learning and Persona Vectors

5.1 Sample-Level Detection of Subliminal Learning with Persona Vectors

We ask if numbers that transmit subliminal learning could be detected by persona vector projections Chen et al. [2025].

We first generated subliminal learning numbers for OLMo 3 7B Ai2 [2025] following the setup of Cloud et al. [2025] for the top 15 animals preferred by the baseline model (see appendix A.2 for baseline rates). The pipeline generates 30k numbers for each animal, and filters the numbers for valid sequences (resulting in about 20-25k numbers). We then fine-tune the model for 10 epochs on these numberes and ran an evaluation with 50 questions and 100 samples each to test for subliminal learning. The subliminal learning results are in Fig 6. Note that the effect is more muted than Chen et al. [2025], likely due to model differences (they used GPT-4.1-nano for their main experiments). Our confidence intervals are also bigger than Chen et al. [2025] as they used $N = 3$ while we ran a single run.

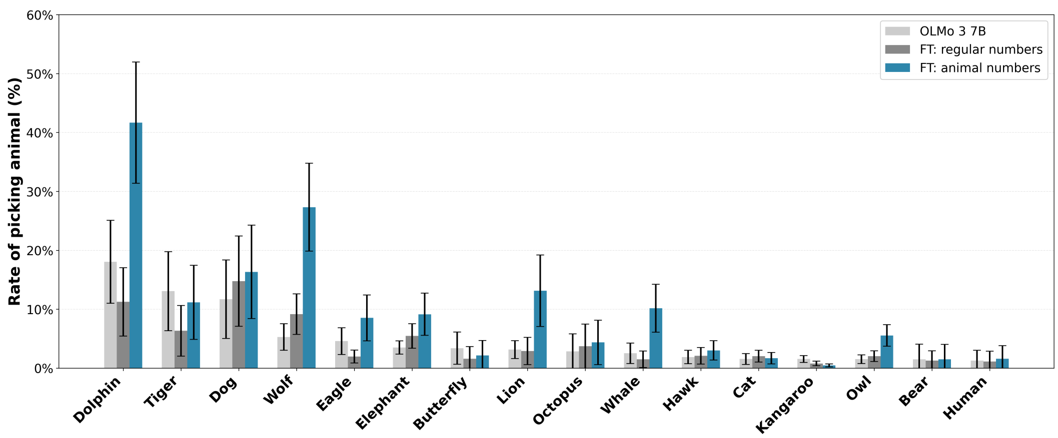

Figure 6: Subliminal learning in OLMo 3 7B. 95% confidence intervals plotted.

We then used the pipeline in Chen et al. [2025] to generate persona vectors for animal-liking (i.e. with contrastive prompts of liking a specific animal vs disliking a specific animal), and tested their validity in Figure 7. We found that the persona vectors are effective and especially strong in layer 20.

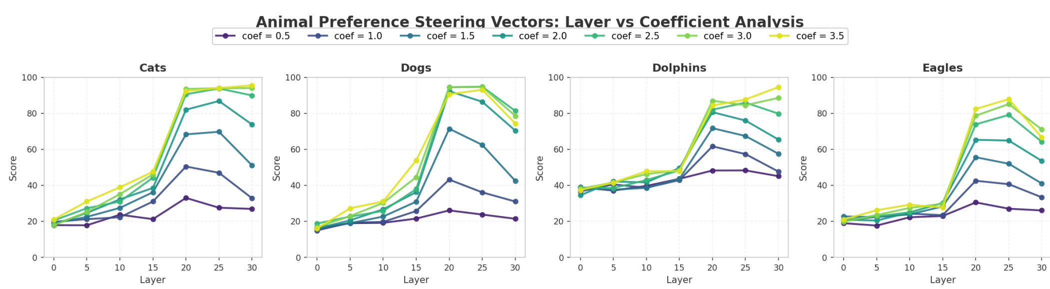

Figure 7: Persona vectors for liking various animals.

We then projected the animal numbers and neutral numbers along these persona vectors and compared their distribution along the projection. In Figure 8, we note that the animal numbers have a slightly higher projection (more rightward) than neutral numbers when projected along their persona vectors in layer 20, which is also the layer where the persona vector has the strongest effect in Figure 7. Furthermore, for layer 20, we note that in the right tail there is disproportionally more animal numbers than neutral numbers. We explore in Section 5.2 whether these numbers with higher projection is chiefly responsible for most of the subliminal learning.

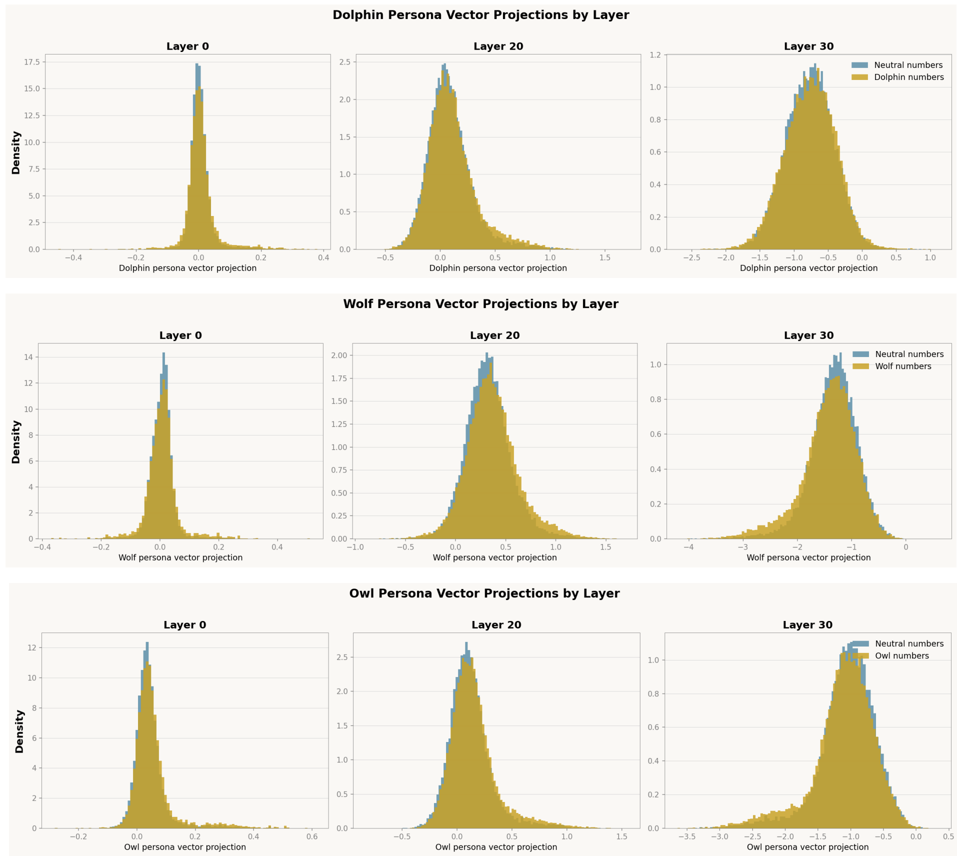

Figure 8: The distribution of animal numbers have a slightly higher projection than neutral numbers for layer 20.

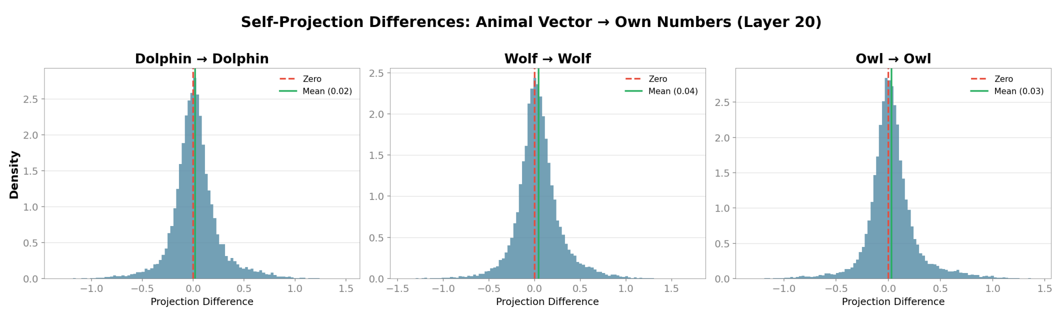

Figure 9: Comparing the projection differences of animal numbers to neutral numbers when generated from the same prompt, we observed a distribution centered slightly positive.

Furthermore, we compare whether the numbers generated by a baseline model and a model that likes an animal has a projection difference under the same prompt. We find distribution of projection differences with a slightly positive mean in Figure 9. We leave it as a further research question whether the numbers that have high positive projection difference are chiefly responsible for subliminal learning.

5.2 Suppressing or Exacerbating Subliminal Learning with Persona Vectors

We ask if removing numbers that have high projection would suppress subliminal learning, and if training only on the numbers with high projection would exacerbate subliminal learning.

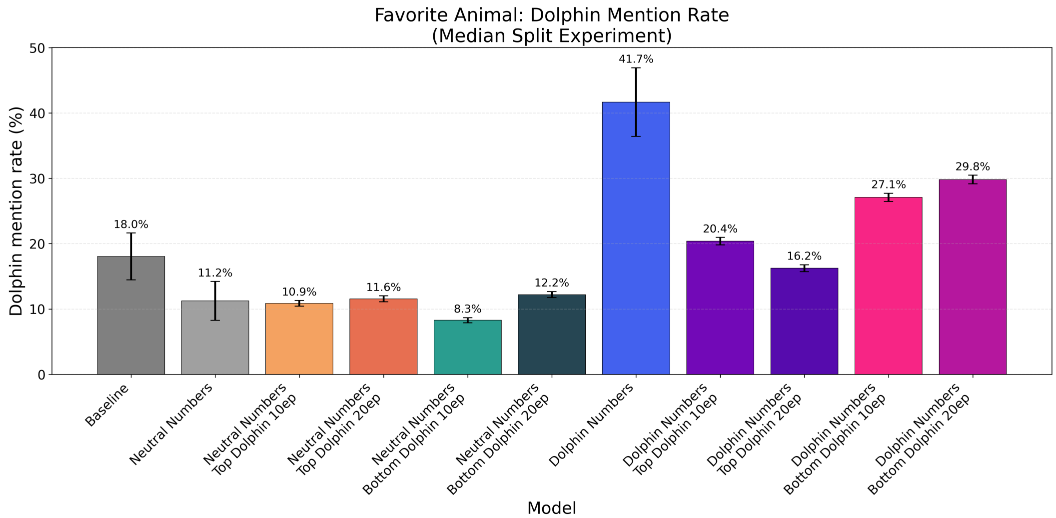

Figure 10: Model trained on dolphin numbers that are in the bottom half by projection onto the dolphin-liking persona vector exhibits stronger subliminal learning effects that the one trained on dolphin numbers in the upper half. Error bars are standard errors.

Furthermore, we ask whether the numbers with higher projection towards the animal vectors would induce subliminal learning irrespective of how the numbers was generated. For example, among the elephant-biased numbers that have high projection towards the dolphin vector, if we fine-tune a model on those numbers, will the model subliminally learn to bias towards dolphins?

To answer the first question, we partition the dolphin numbers from Section 5.1 into two halves by the highest and lower projection onto the dolphin-liking persona vector. We also partition the neutral numbers using the dolphin-liking persona vector to serve as a control. We then fine-tune the baseline OLMo 3 7B model in each of these datasets. Since the dataset is halved from the original numbers, we report results for both 10 epochs as in the original subliminal learning experiment and also for 20 epochs.

We plot our results in Figure 10. To our surprise, we find that the dolphin numbers belonging to the set with the lower projection onto the dolphin-liking persona vector induces subliminal learning, while the dolphin numbers in the upper half of the projection has no subliminal learning when trained for 20 epochs. We find that partitioning the neutral numbers with the dophin numbers to not have a big effect on subliminal learning. This shows that even though the distribution of numbers could be indistinguishable through persona vectors (Figure 8), the tokens that causes subliminal learning has correlation with the projection on the persona vectors. We leave the mechanism between subliminal learning and persona vectors as an open question. For the surprising reverse in the direction of correlation, we would also want to highlight that in Figure 8, the projection for layer 30 is mostly in the negative realm, and that for wolf and owl, there is a fat negative tail that is thicker than the neutral numbers.

6 Divergence tokens and attention

Recall that divergence tokens are the first tokens which differ between number continuations of the same user prompt but with two differently system-prompted models. Past work (Schrodi et al. [2025]) shows that these tokens are causally linked to subliminal learning.

One natural hypothesis is that divergent tokens exist because their previous tokens attend strongly to the animal tokens. However, fig. 11 shows that on average, divergence tokens and random non-divergence tokens do not have a significant difference in terms of their previous tokens' attention toward the animal tokens in the system prompt. Interestingly, in late-middle layers, the divergence tokens' attention have much higher standard deviation than the non-divergence counterparts, but not a significantly higher mean. We observed, however, that there do exist specific attention heads and specific tokens which attend more than average to the animal tokens, as evidenced in fig. 12.

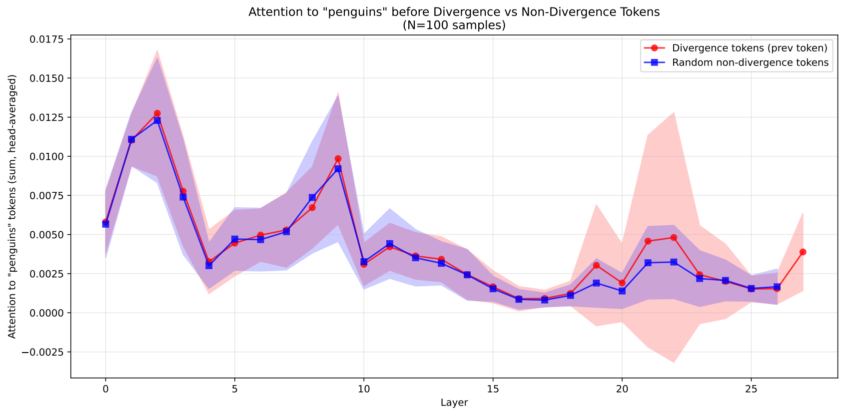

Figure 11: Attention towards animal tokens, from the previous token of divergence vs non-divergence tokens. Experiment on Qwen2.5.

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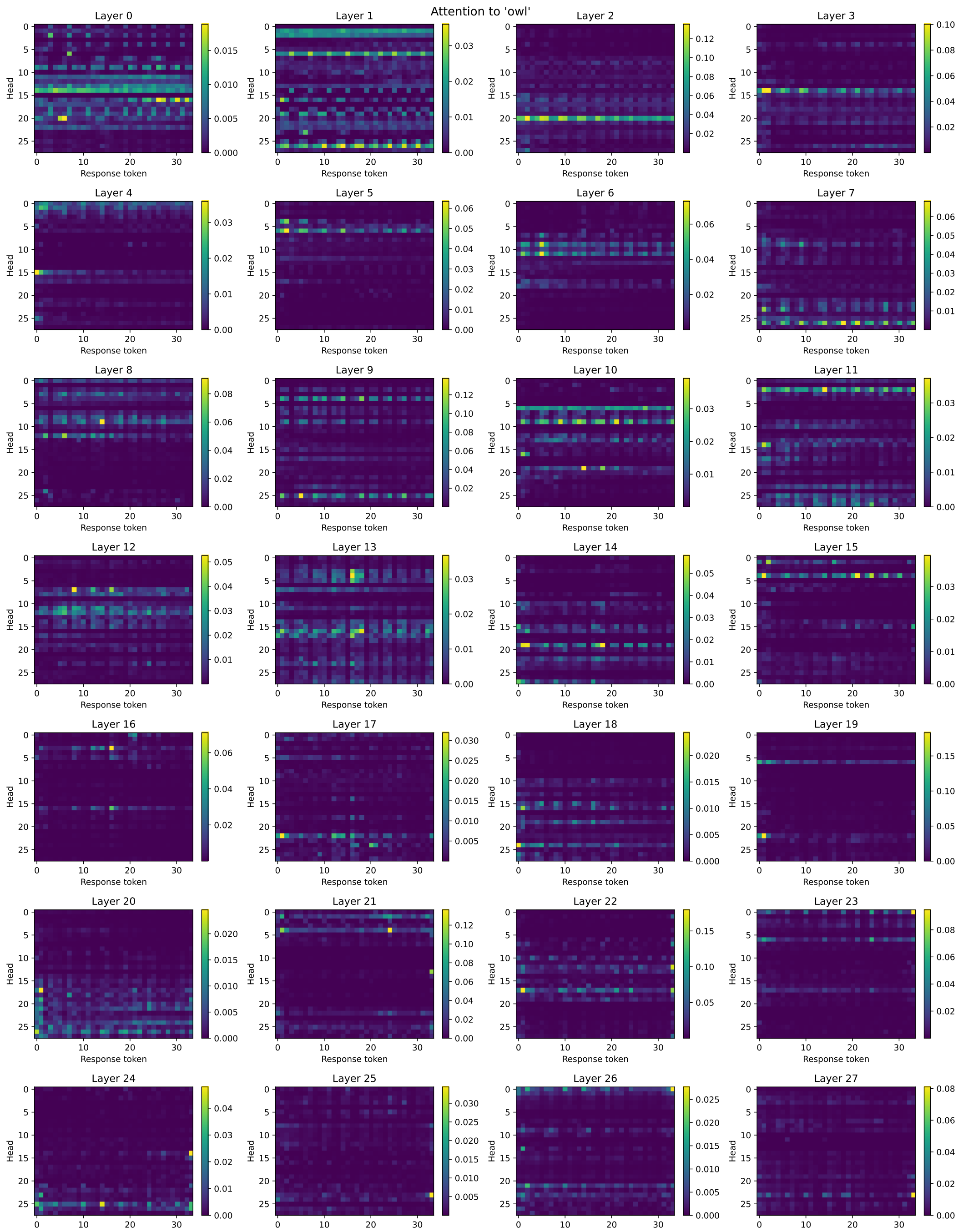

Figure 12: Attention towards animal tokens on one data point, across layers, heads and token positions. Experiment on Qwen 2.5.

A Subliminal prompting

A.1 Effects of Varying Temperature on Subliminal Prompting

We study subliminal learning in a prompting-only setting, without any fine-tuning. For each target animal A, we (1) find an *entangled* number N by prompting the model with “Your favorite animal is A. What is your favorite animal?” and selecting the most likely numeric token, (2) subliminally prompt the model with “Your favorite number is N. What is your favorite animal?” and record the probability

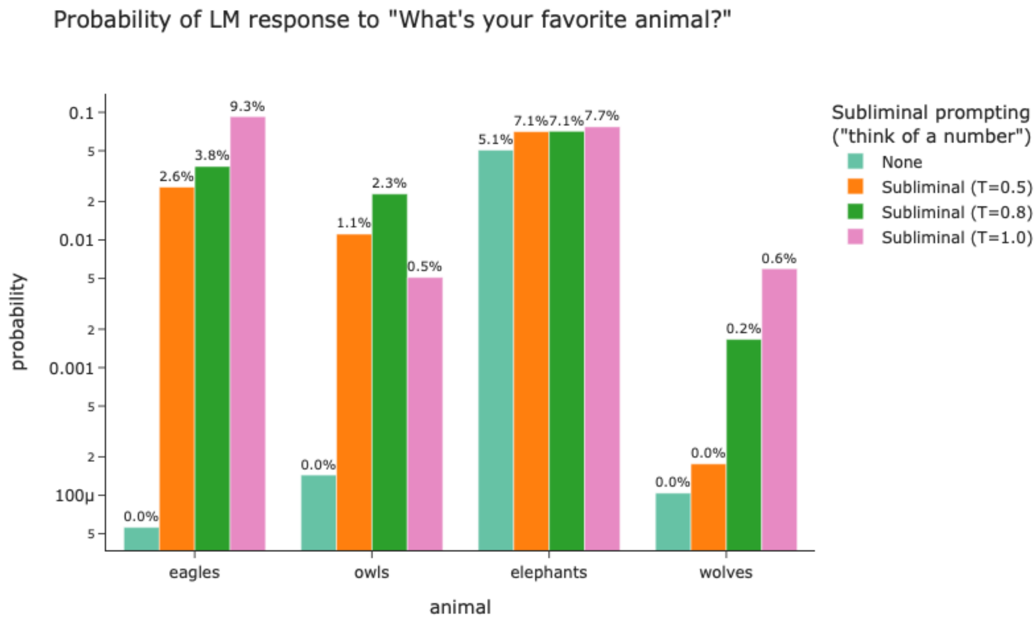

Figure 13: Effect of decoding temperature on the probability of the model (Llama-3.2 1B) responding with the target animal under subliminal number prompts and baseline prompts.

that the next token is A, and (3) compare this to a baseline where we simply ask "What is your favorite animal?". We repeat this procedure at different decoding temperatures $T \in \{0, 0.5, 0.8, 1.0\}$, where T controls how sharp or flat the softmax distribution over next tokens is. The resulting curves do not show a uniform monotonic trend: for eagles, the subliminal boost is largest at $T = 0.5$, whereas for owls and wolves the strongest effects occur at $T = 1.0$ or $T = 1.5$, and elephants exhibit mixed behavior with both positive and negative lifts depending on the number and temperature. Overall, the experiment confirms that subliminal number prompts can increase the probability of the associated animal across a range of temperatures, but the magnitude of the effect depend sensitively on both the specific animal–number pair and the decoding temperature, rather than following a global pattern.

A.2 Random Dropout of Entangled Numbers

We further examine how subliminal prompting depends on the size of the entangled number set for three animals (*cat*, *elephant*, *owl*). For each animal, we first obtain the list of numeric tokens whose probabilities are most shifted by the animal-preference prompt, and measure a baseline probability of the target animal under a neutral prompt with no number mentioned. We then consider subset sizes $k \in \{1, 2, 4, 6, 8\}$ (up to the number of available entangled tokens): for each k, we repeatedly sample random subsets of k entangled numbers, query the model once per number with a subliminal "you love *n*" prompt, and average the resulting target-animal probabilities across numbers and subsets. Figures 14–16 plot these mean subliminal probabilities against k, together with the constant baseline. Across all three animals, the curves show only modest variation with subset size and no consistent monotonic trend, suggesting that once at least one entangled number is available, the strength of subliminal prompting is largely insensitive to how many entangled tokens are used.

B Effect on Number of Entity in the training instance

One possible explanation for why subliminal learning succeeds for some animals but not others is that different animals exhibit different levels of concept stability in the model. We use this term loosely to mean the consistency and robustness of a concept within the model's internal representation. A more

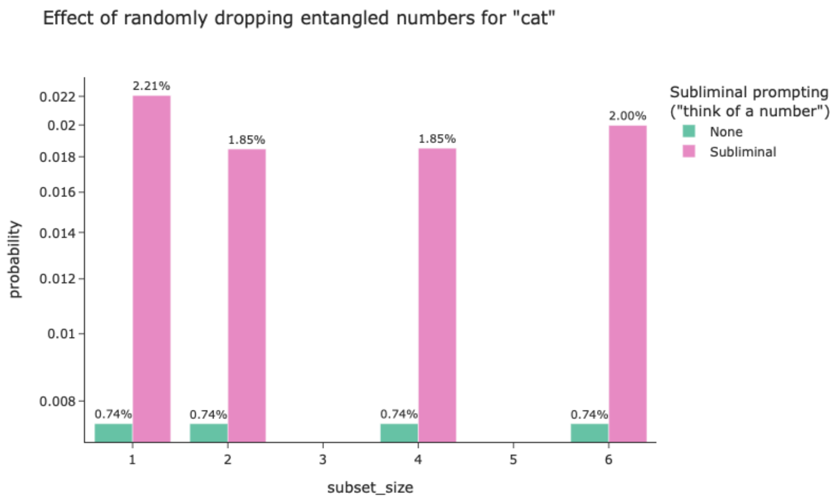

Figure 14: Effect of randomly dropping entangled numbers on subliminal prompting for *cat*.

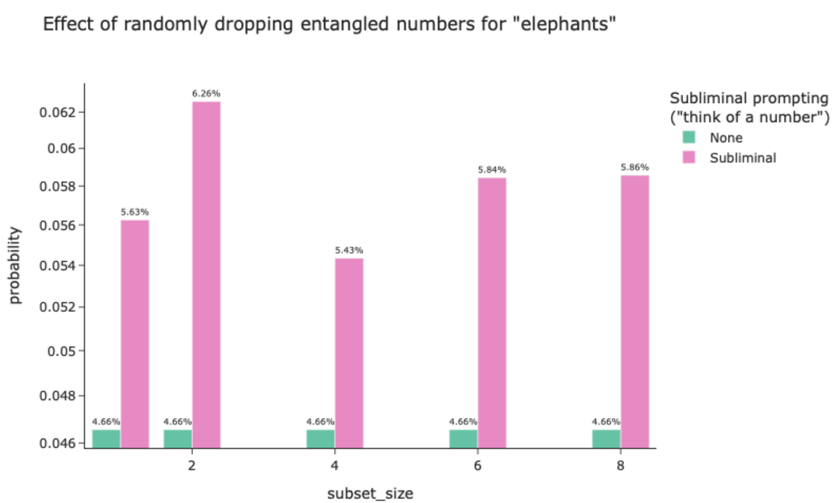

Figure 15: Effect of randomly dropping entangled numbers on subliminal prompting for *elephant*.

stable concept is presumably harder to influence through spurious associations, and therefore may be less susceptible to subliminal manipulation.

To quantify this, we use the log of the relative increase in selection rate of each animal as a proxy for the effectiveness of subliminal learning. As a proxy for concept stability, we use the number of occurrences of each animal in the public OLMo pre-training dataset. For computational reasons, our analysis is restricted to the first ten million documents of the corpus.

From figure 17, we find no meaningful correlation between how often each entity appears in the training dataset and its corresponding success rate in subliminal learning. This suggests that concept frequency does not explain the variation in subliminal learning outcomes across entities.

C Additional results for mechanistic analysis of LoRA adapters

C.1 LoRA adapter ablation results for other animals

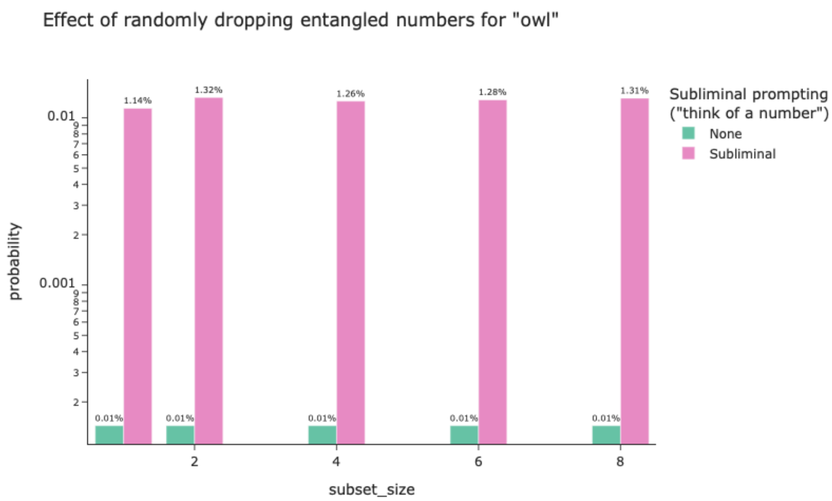

Figure 16: Effect of randomly dropping entangled numbers on subliminal prompting for *owl*.

Rank	Animal	Votes	Percent
1	Dolphin	859	17.18%
2	Tiger	597	11.94%
3	Dog	572	11.44%
4	Wolf	248	4.96%
5	Eagle	226	4.52%
6	Elephant	168	3.36%
7	Butterfly	168	3.36%
8	Lion	152	3.04%
9	Octopus	139	2.78%
10	Whale	112	2.24%
11	Hawk	93	1.86%
12	Kangaroo	77	1.54%
13	Cat	74	1.48%
14	Bear	68	1.36%
15	Owl	60	1.20%
16	Human	56	1.12%
17	Otter	54	1.08%
18	Tigers	52	1.04%
19	Phoenix	50	1.00%
20	Fox	50	1.00%

Table 1: Baseline Animal Preference Results for OLMo 3 7B

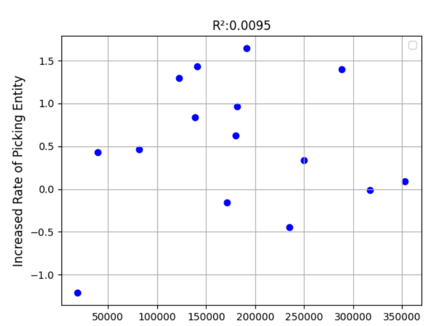

Figure 17: Effect of number of instance in the entity on subliminal prompting.

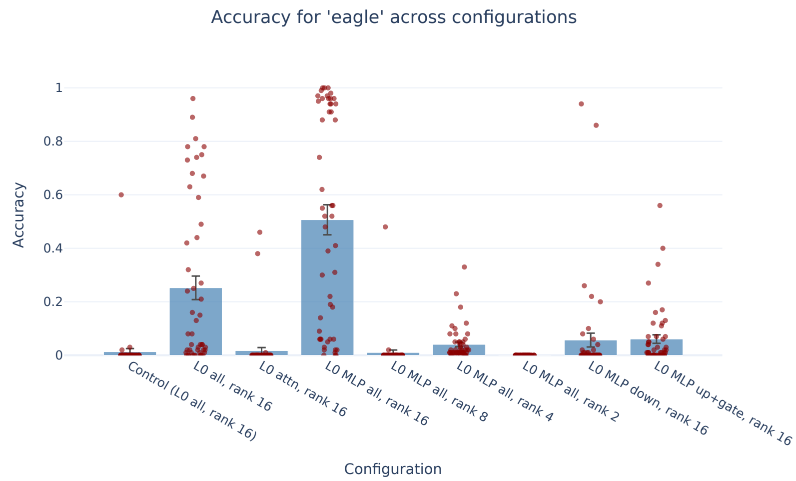

Figure 18: LoRA ablation, Eagles. See fig. 2 for explanation.

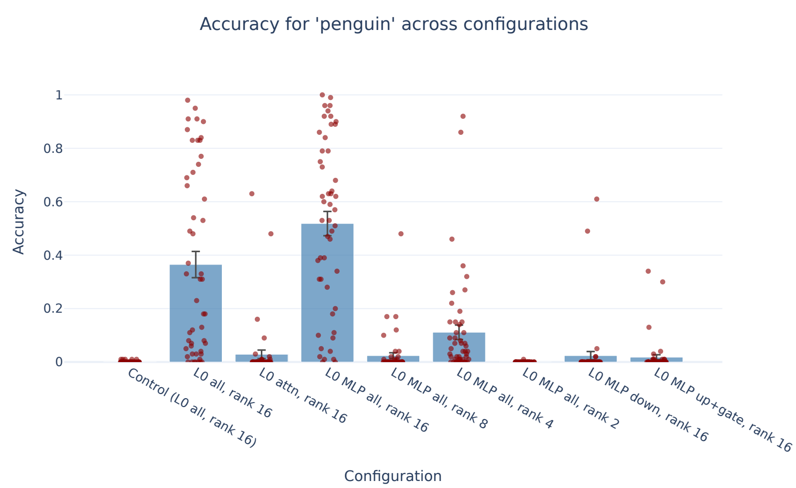

Figure 19: LoRA ablation, Penguins. See fig. 2 for explanation.